MeerKAT Galactic Center

Data Reduction Summary

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1. Observation

We use data from the MeerKAT Galactic Centre Legacy Survey (proposal SCI-20210212-SS-01). The observation targets Sgr A* in L-band (856-1712 MHz) with 32,768 spectral channels at a native channel width of 26.123 kHz. The total measurement set (MS) is ~17 TB. The data were pre-calibrated by the MeerKAT Science Data Processor (SDP) pipeline prior to archive delivery.

2. Subband Splitting

The monolithic MS was split into 86 independent subbands of 383 channels each (~10 MHz per subband) using the CASA split task (casatasks.split, CASA 6.x). Splitting was performed serially to avoid file-locking contention on the Lustre parallel filesystem. Each split operation selected a contiguous range of channels via the spw parameter (spw="0:start~end") and wrote the DATA column to a new MS. The typical split time was ~90 minutes per subband. No frequency regridding was applied; the TOPO-to-LSRK Doppler correction is <0.1 channel widths for a single observation and is applied as a metadata correction in post-processing.

3. Imaging

Channel images were produced using the CASA tclean task in cube mode (specmode="cube"), generating a 383-channel image cube per subband in a single tclean call. Each cube was then decomposed into individual single-channel FITS files using imsubimage and exportfits, yielding one 512x512 pixel image per spectral channel at 2 arcsec pixel scale (between 1/3 and 1/5 of the synthesized beam across the band). Briggs weighting with robust=0.5 was used. The full dataset produces 32,768 channel images spanning 856-1712 MHz at the native 26.123 kHz resolution.

Dirty images (niter=0) are the direct Fourier inversion of the calibrated visibilities with no deconvolution, providing an unbiased representation of the sky convolved with the synthesized beam (PSF). The dirty image noise is ~0.09 Jy/beam per 26 kHz channel in the mid-band (~1.1 GHz).

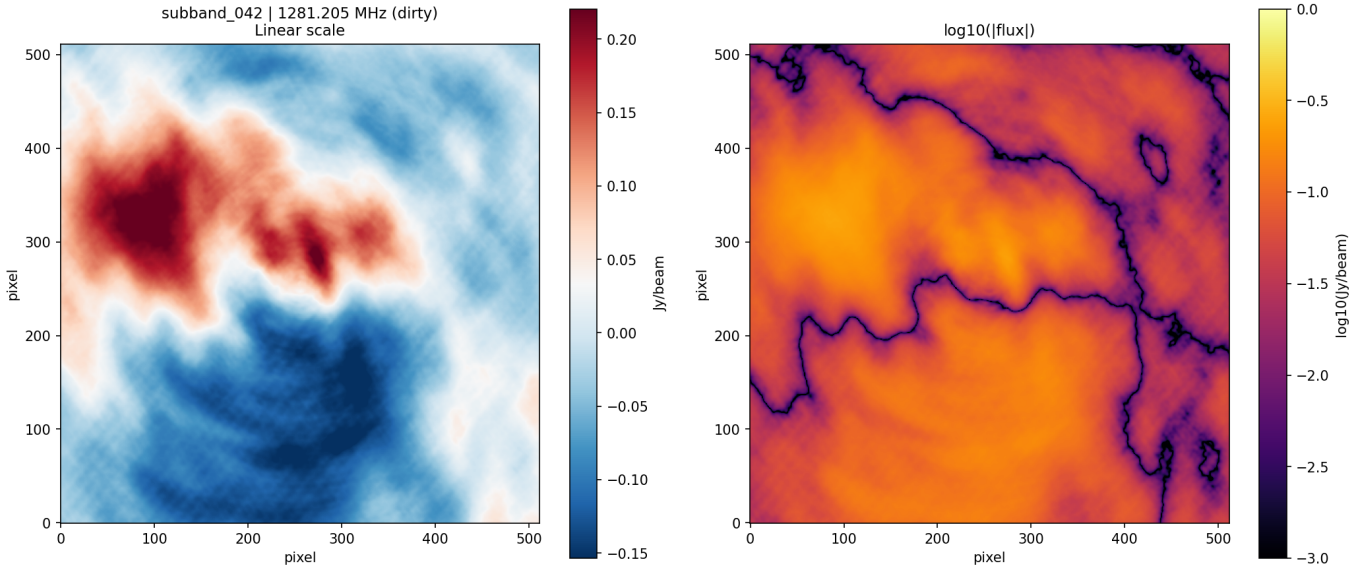

Figure 1: Representative dirty channel image at ~1.1 GHz. Left: linear scale showing the thermal noise and source structure. Right: log scale revealing faint emission and PSF sidelobes across the field.

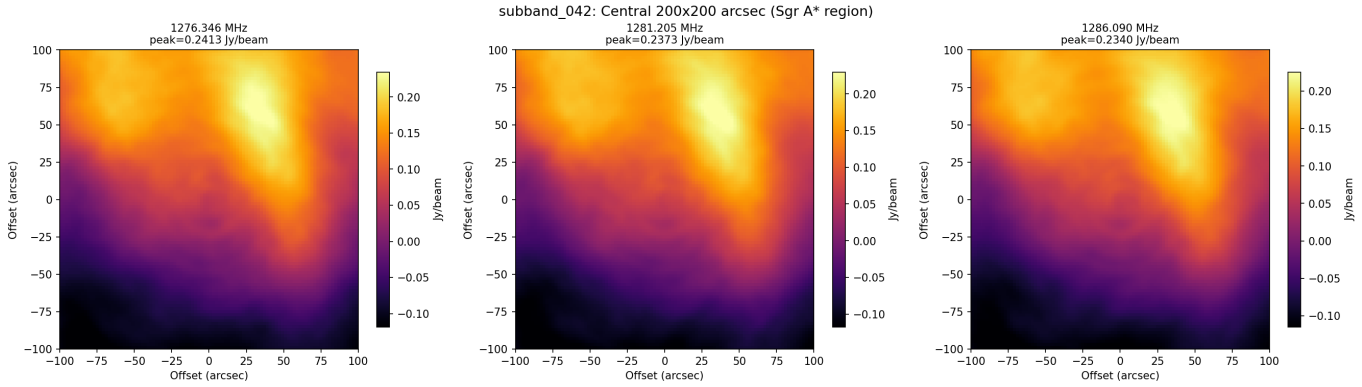

Figure 2: Zoomed view of the central 200x200 arcsec (Sgr A region) at three frequencies within one subband. The dominant compact source is clearly detected in each 26 kHz channel.*

4. Computational Resources

All processing was performed on the Lawrencium HPC cluster at UC Berkeley, managed by SLURM. Subband splitting was submitted as a serial array job (max 8 concurrent tasks to match the 8 Lustre OSTs over which the MS is striped). Imaging was fully parallelized as a SLURM array job with one subband per compute node (56 cores, 240 GB RAM per node, lr7 partition). Dirty imaging required ~50 minutes per subband wall-clock time.

5. Data Products

The primary data products are individual FITS images for each of the 32,768 spectral channels. Each FITS file contains a single Stokes I image of 512x512 pixels at 2 arcsec resolution, in units of Jy/beam. Channel frequencies are encoded in the FITS filename (chan_NNNN_FFFF.FFFMHz.fits) using the exact channel frequency from the MS metadata.

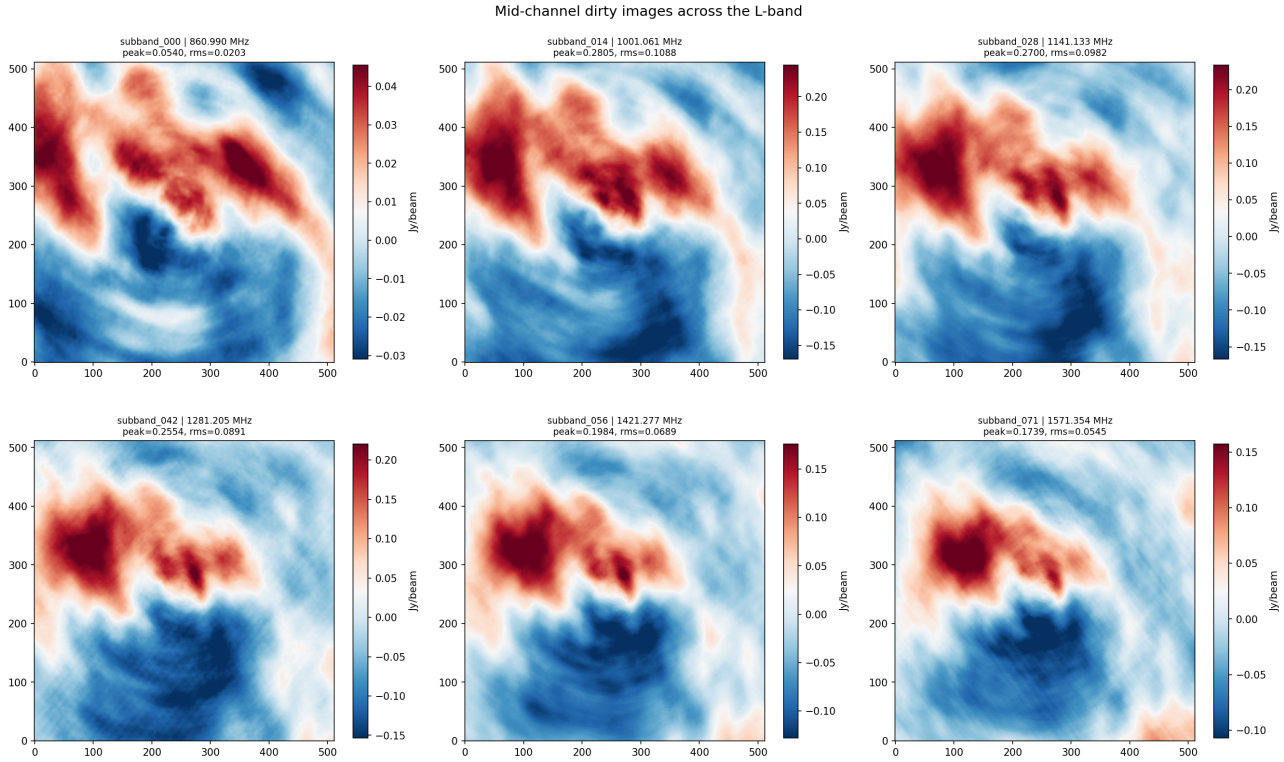

Figure 3: Mid-channel dirty images from six subbands spanning the L-band. The source morphology evolves with frequency as expected from the frequency-dependent synthesized beam and sky brightness distribution.

6. Frequency Coverage and RFI

The MeerKAT L-band receiver covers 856-1712 MHz. Several known radio frequency interference (RFI) sources affect portions of the band, including GSM-900 downlink (925-960 MHz), aircraft DME/SSR (1030-1090 MHz), GPS L5/Galileo E5 (1164-1214 MHz), GPS L2 (1217-1237 MHz), GPS L1/Galileo E1 (1559-1591 MHz), Glonass L1 (1598-1610 MHz), and Iridium (1610-1618 MHz). No RFI flagging has been applied to the current images; the data were delivered with the MeerKAT SDP calibration but without post-delivery RFI excision. The relatively RFI-quiet region around 1.3 GHz is of primary interest for the axion search.

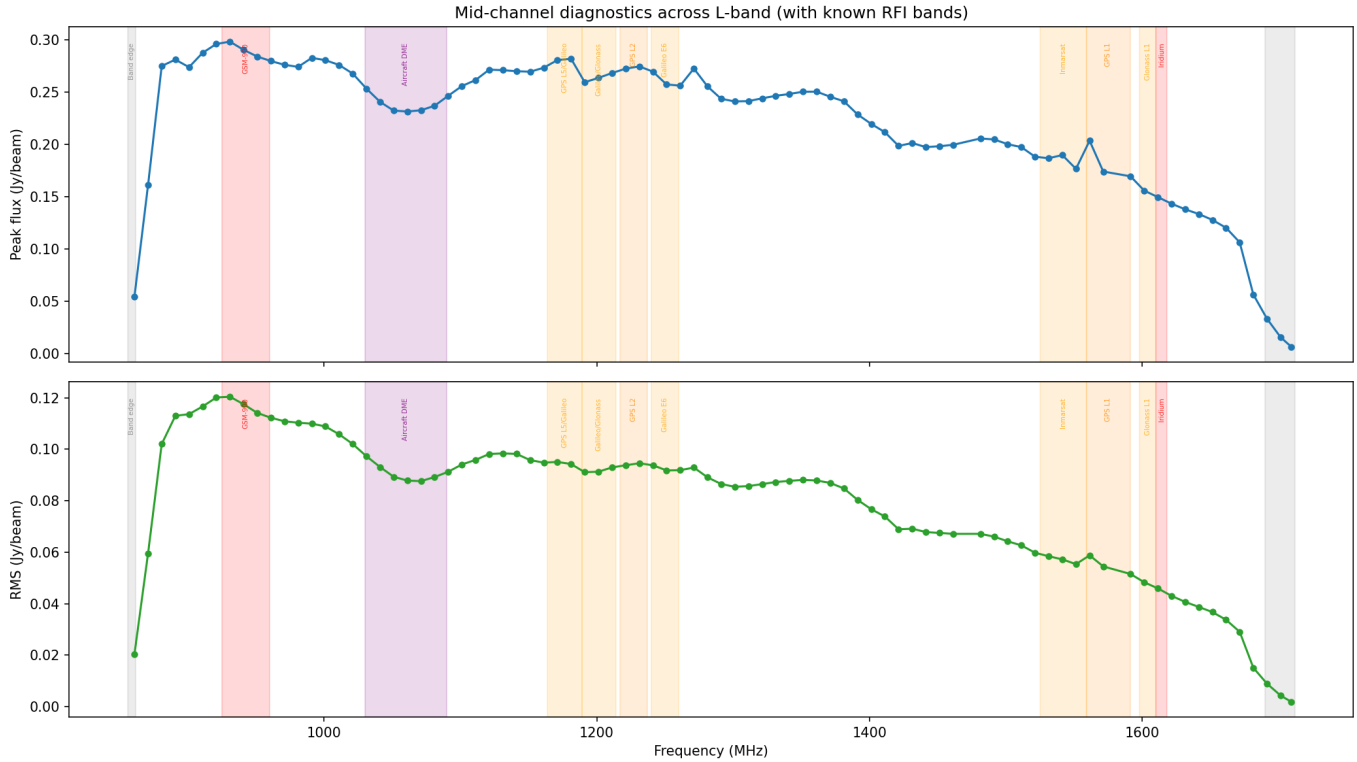

Figure 4: Peak flux density (top) and RMS noise (bottom) of the mid-channel dirty image for each completed subband across the L-band. Shaded regions indicate known MeerKAT RFI bands. Elevated noise is visible near the GSM-900 and aircraft DME bands.

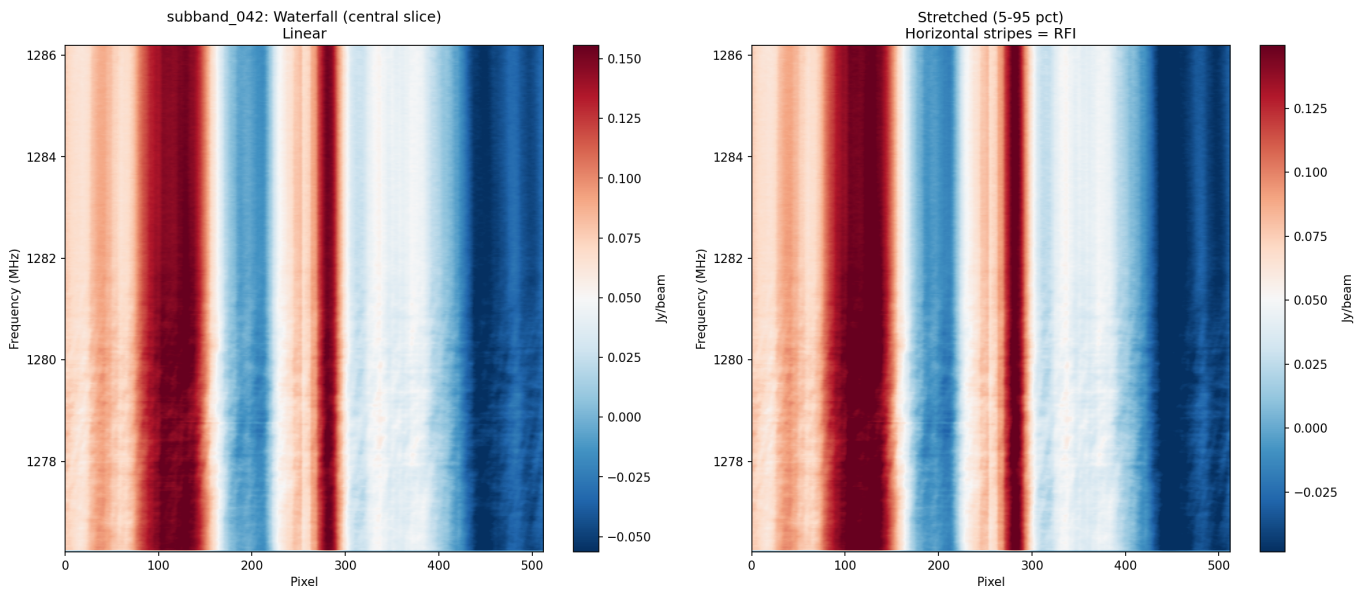

Figure 5: Waterfall plot showing flux density along a horizontal slice through the image center as a function of frequency for one subband. Left: full dynamic range. Right: stretched scale (5th-95th percentile). Horizontal stripes would indicate RFI-affected channels; the smooth vertical structure traces the central source.

7. Known Systematics

The first channel of each subband exhibits elevated peak flux (approximately 3x the median), consistent with a bandpass edge effect introduced by the channel splitting. These edge channels are excluded from the science analysis. The FITS WCS frequency axis (CRVAL3) records the tclean cube reference frequency rather than the per-channel frequency; the true channel frequency is obtained from the filename or computed via CRVAL3 + (1 - CRPIX3) * CDELT3.

8. Image Validation

Several diagnostic checks were performed on the dirty images to verify data quality and pipeline correctness.

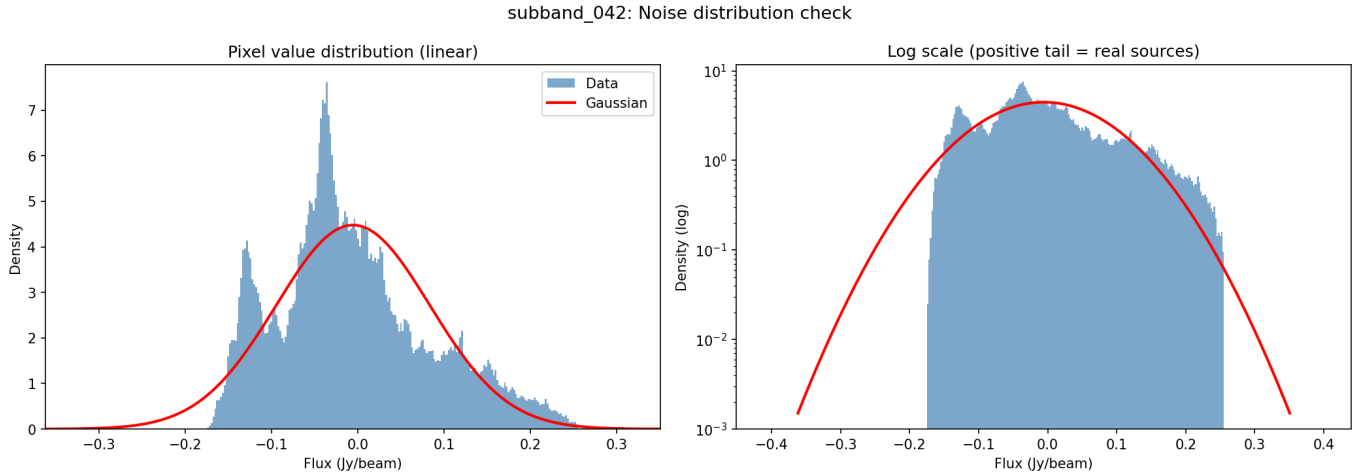

Figure 6: Pixel value distribution for a representative mid-band channel compared to a Gaussian fit. Left: linear scale showing the noise core is well-described by thermal noise. Right: log scale revealing the positive tail from real astronomical emission. The Gaussian noise core confirms proper calibration.

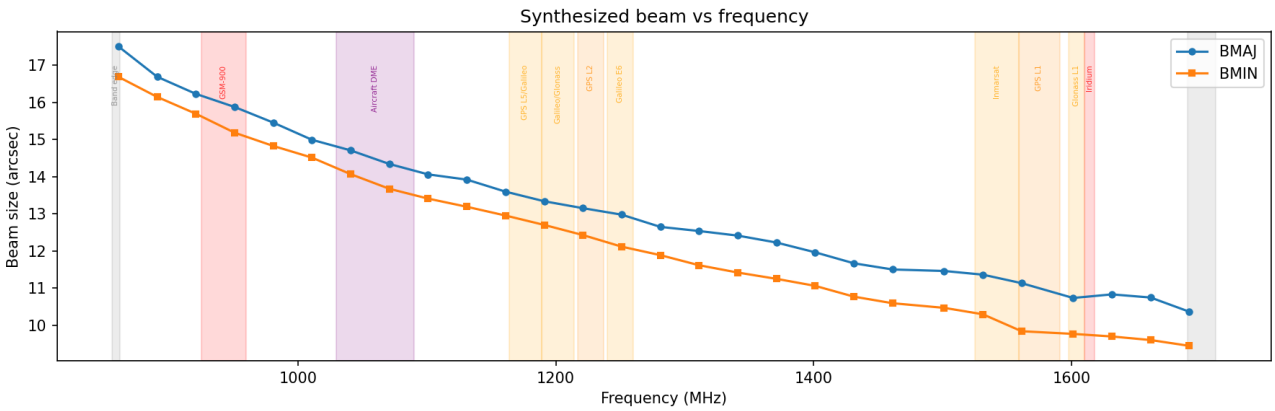

Figure 7: Synthesized beam major and minor axis as a function of frequency. The beam size decreases at higher frequencies as expected (beam ~ wavelength/baseline). Shaded bands indicate known RFI regions.

9. Science Application

The native 26.123 kHz channel resolution is preserved throughout the imaging pipeline, as required for the axion-photon conversion search. The expected axion signal line width may be comparable to the channel width, making frequency downbinning unacceptable. Each channel image will be analyzed independently for narrow spectral features consistent with axion dark matter conversion in neutron star magnetospheres across the 856-1712 MHz band (corresponding to axion masses of ~3.5-7.1 ueV).

Summary Table

Telescope	MeerKAT (64 dishes)
Band	L-band, 856-1712 MHz
Channels	32,768 @ 26.123 kHz
Target	Sgr A* / Galactic Center
Calibration	MeerKAT SDP pipeline (pre-calibrated)
Image size	512 x 512 pixels, 2 arcsec/pixel
Weighting	Briggs, robust = 0.5
Dirty image noise	~0.09 Jy/beam per channel (mid-band)
Data products	32,768 FITS channel images
Total MS size	~17 TB
Processing cluster	Lawrencium (UC Berkeley HPC)
Time per subband	~50 min (dirty imaging)

10. Software

CASA 6.x (modular installation via pip: casatools, casatasks). Image analysis and visualization: astropy, matplotlib, numpy. Cluster job management: SLURM. All scripts are available in the project repository under scripts/.